Networking Tools & Techniques

Unit-I: Introduction to Networking

Network: A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

Networking is the process of making connections and building relationships.

Computer Network: A computer network is a system that connects two or more computing devices for transmitting and sharing information. Computing devices include everything from a mobile phone to a server. These devices are connected using physical wires such as fiber optics, but they can also be wireless.

Goals of Computer Networks: The following are some important goals of computer networks:

1. **Resource Sharing –** Many organization has a substantial number of computers in operations, which are located apart. Ex. A group of office workers can share a common printer, fax, modem, scanner, etc.

2. **High Reliability –** If there are alternate sources of supply, all files could be replicated on two or more machines. If one of them is not available, due to hardware failure, the other copies could be used.

3. **Inter-process Communication –** Network users, located geographically apart, may converse in an interactive session through the network. In order to permit this, the network must provide almost error-free communications.

4. **Flexible access –** Files can be accessed from any computer in the network. The project can be begun on one computer and finished on another.
5. **Security**– Computer networks must be secure to protect against unauthorized access, data breaches, and other security threats. This includes implementing measures such as firewalls, antivirus software, and encryption to ensure the confidentiality, integrity, and availability of data.
6. **Performance**– Computer networks must provide high performance and low latency to ensure that applications and services are responsive and available when needed. This requires optimizing network infrastructure, bandwidth utilization, and traffic management.
7. **Scalability-** Computer networks must be designed to scale up or down as needed to accommodate changes in the number of users, devices, and data traffic. This requires careful planning and management to ensure the network can meet current and future needs.
   Other goals include Distribution of processing functions, Centralized management, and allocation of network resources, Compatibility of dissimilar equipment and software, Good network performance, Scalability, Saving money, Access to remote information, Person to person communication, etc.

Types of Network Applications

There are several types of network applications, each designed to serve specific purposes and meet diverse communication and data-sharing needs:

1. **Web Browsers:**

- Examples: Google Chrome, Mozilla Firefox, Microsoft Edge
- Web browsers allow users to access and navigate websites and web-based applications over the internet. They use network protocols such as HTTP and HTTPS to retrieve web content.

2. **Email Clients:**

- Examples: Microsoft Outlook, Apple Mail, Gmail
- Email clients facilitate the sending, receiving, and management of email messages over email servers, typically using protocols like SMTP, IMAP, and POP3.

3. **File Transfer Protocols:**

- Examples: FTP (File Transfer Protocol), SFTP (Secure File Transfer Protocol)
- These applications enable the transfer of files between devices and servers. FTP and SFTP ensure secure and efficient file exchange over networks.

4. **Messaging Apps:**

- Examples: WhatsApp, Slack, Microsoft Teams
- Messaging applications allow real-time text, voice, and video communication between individuals and groups, enhancing collaboration and connectivity.

5. **Video Conferencing Tools:**

- Examples: Zoom, Microsoft Teams, Cisco Webex
- Video conferencing applications enable virtual meetings, webinars, and remote collaboration through live video and audio communication.

6. **Remote Desktop Applications:**

- Examples: TeamViewer, AnyDesk, Remote Desktop Protocol (RDP)
- These applications allow users to access and control remote computers over a network, facilitating technical support, troubleshooting, and remote work.

Importance of Network Applications

Network applications play a pivotal role in today's interconnected world for several reasons:

1. Enhanced Connectivity:

- Network applications enable individuals and organizations to connect and communicate seamlessly, bridging geographical distances and fostering collaboration.

2. Data Sharing and Accessibility:

- They facilitate the sharing and access of data and resources, promoting efficient information exchange and decision-making.

3. Streamlined Workflow:

- Network applications automate processes, reducing manual tasks and enhancing productivity across various industries.

4. Improved Communication:

- These applications offer real-time communication channels, supporting video conferencing, instant messaging, and email, which are essential for business operations and personal interactions.

5. Remote Work and Flexibility:

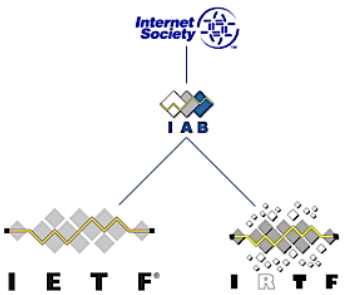
- Network applications empower remote work by providing secure access to corporate resources, enabling businesses to adapt to changing work environments.

6. Scalability and Growth:

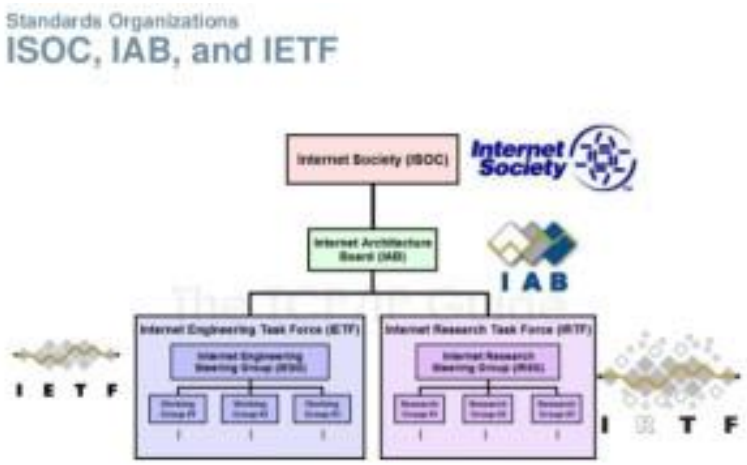
- As businesses expand, network applications can scale to accommodate increased demands, ensuring they remain effective tools for communication and collaboration.

Organization of the Internet

The Internet is a collection of autonomous and interconnected networks that implement open protocols and standards. No single person, government or organisation owns or controls the Internet or the World Wide Web. There are, however, a number of organisations and bodies – both international and national – which regulate, monitor, make recommendations or set guidelines covering issues such as infrastructure, domain registration, web technologies, content and accessibility.



A non-profit organization called **ISOC** (**Internet Society**) has some control over the future of the Internet. It appoints a technical advisory group called the **IAB** (**Internet Architecture Board**) to evaluate and set standards.
Input on protocols and standards can come from anybody – individuals, research groups, companies and universities. A specification is submitted as an Internet draft and made available for review and comments. Various Internet organizations evaluate whether these specifications should be advanced – through a process that may elevate the specification through different levels until it potentially reaches a 'standards' status.



In general, the **IETF** (**Internet Engineering Task Force**) forms working groups to develop specifications, which are evaluated by the **IESG** (**Internet Engineering Steering Group**) in conjunction with the **IAB**. **ISOC** then publicizes the new standards. Web standards are promulgated by the **W3C** (**World Wide Web Consortium**) and other groups.

Internet Service Provider (ISP)

ISP stands for **Internet Service Provider** which is a term used to refer to a company that provides internet access to people who pay the company or subscribe to the company for the same. For their services, the customers have to pay the internet service provider a nominal fee which varies according to the amount of data they actually use or the data plan which they purchase. An Internet Service Provider is also known as an Internet Access Provider or an online service provider. An Internet Service Provider is a must if one wants to connect to the internet.

History

The first Internet Service Provider was Telenet. Telenet was the commercialized version of the ARPANET – a precursor to the internet, of sorts. Telenet was introduced in 1974. Since then, many Internet Service Providers have entered the scene and this was partly because of the proliferation of the internet as a commodity that fuelled the consumerist attitude of the people. Pretty soon, an Internet Service Provider called "The World" came to be in vogue and ever since it started serving its customers today in 1989 has cemented itself as the first archetypal Internet Service Provider. Examples of major Internet Service Providers include Google Fiber, Verizon, Jio, AT&T etc.

Characteristics

- **E-mail Account:** Many Internet Service Providers offer an e-mail address to their consumers.
- **User Support:** Professionals and an increasing number of lay users prefer an ISP that can provide them with customer support so that they have someone they can refer to if things go awry.
- **Access to high-speed internet:** Probably the most obvious item on this list as this feature of an Internet Service Provider lies literally in its name. Furthermore, the higher the speed an Internet Service Provider can offer one, the better it's standing in the market and the more customers it can attract.
- **Spam Blocker:** An Internet Service Provider that hinders its customers' productivity by way of not blocking spam and displaying frequent ads is not something that is generally favoured in the market today. Therefore, many of the Internet Service Providers offer spam blocking features to their customers.
- **Web Hosting:** Some of the ISPs offer web hosting services to their clientele as well.

Different types of ISP connections

- DSL
- Wi-Fi broadband
- mobile broadband
- fibre optic broadband
- cable broadband

List of ISP

- Reliance Jio
- Vodafone Idea
- Airtel
- BSNL
- Hathway

Advantages

- The customer need not then bother with either the technicalities or finances of investing and inventing a web browser to work with. An ISP can readily do all of this for its customers.
- Many ISPs, being professional companies, provide its clientele with high-speed internet and that is not possible if one decides to sidesteps these companies.
- ISPs offer a very high degree of reliability and availability
- The ISPs are secure – they offer a tremendous deal of protection against viruses and use only the latest software patches whilst operating and thereby, maintaining the integrity of the browser.
- User do not need to invest in user's own web server.
- ISP's should give the best uptime guarantee.

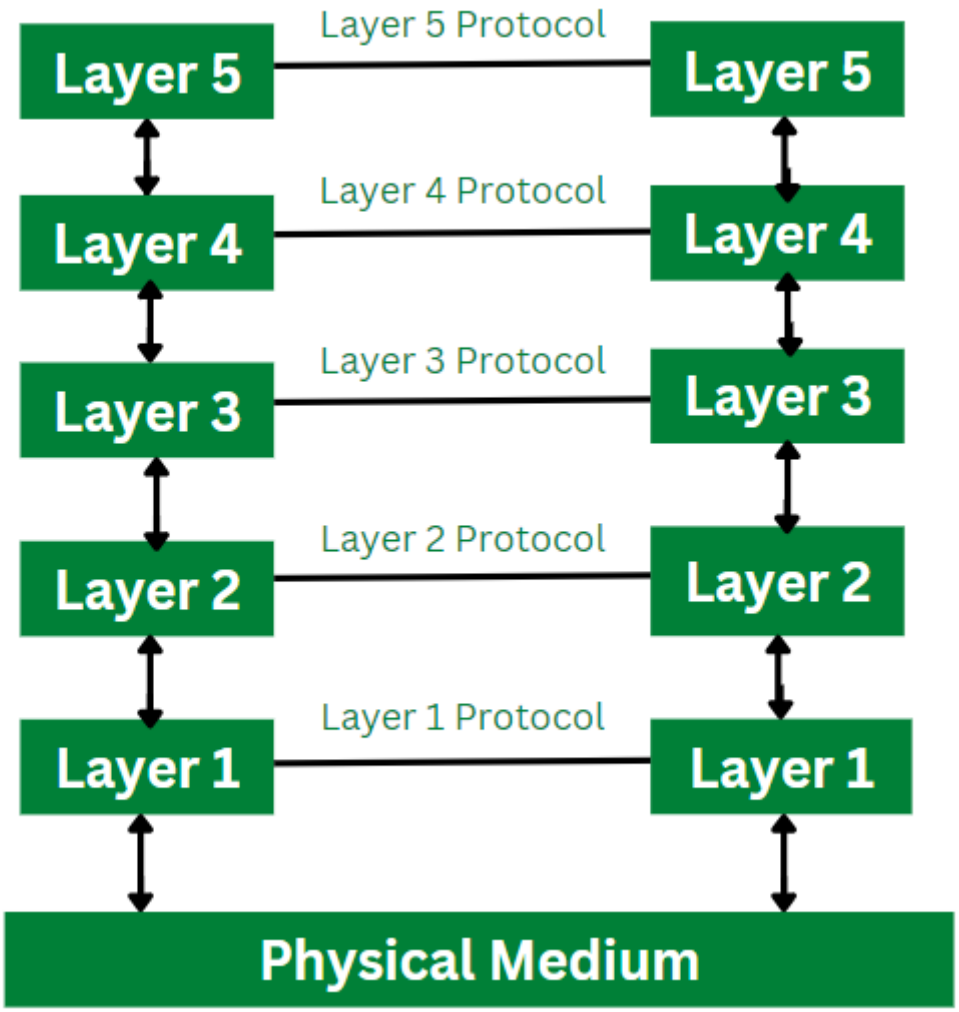
Disadvantages

- Because of the range of options available in the market and due to cut-throat competition, some of the ISPs have been accused of violating the customers' trust by way of inflated pricing, data losses, etc. It is true that using an ISP makes the customer entirely dependent on it.

- If an Internet Service Provider is stretched thin because of hosting too many sites on a shared server, it can compromise the quality of the customers' data by way of slow download rates and poor performance of websites.
- User need to trust user's ISP for uptime and security.
- ISP can directly affect user if the it gets blacklisted.

❖ **Network Structure and Architecture**

Every network consists of a specific number of functions, layers, and tasks to perform. Layered Architecture in a computer network is defined as a model where a whole network process is divided into various smaller sub-tasks. These divided sub-tasks are then assigned to a specific layer to perform only the dedicated tasks. A single layer performs only s specific type of task. To run the application and provide all types of services to clients a lower layer adds its services to the higher layer present above it. Therefore layered architecture provides interactions between the sub-systems. If any type of modification is done in one layer it does not affect the next layer.



Layered Architecture

As shown in the above diagram, there are five different layers. Therefore it is a five-layered architecture. Each layer performs a dedicated task. The lower level data for example from layer 1 data is transferred to layer 2. Below all the layers Physical Medium is present. The physical medium is responsible for the actual communication to take place. For the transfer of data and communication layered architecture provides with a clean cut interface.

Features of Layered Architecture

- Use of Layered architecture in computer network provides with the feature of modularity and distinct interfaces.
- Layered architecture ensures independence between layers, by offering services to higher layers from the lower layers and without specifying how this services are implemented.
- Layered architecture segments as larger and unmanageable design into small sub tasks.
- In layer architecture every network has different number of functions, layers and content.
- In layered architecture, the physical route provides with communication which is available under the layer 1.

- In layered architecture, the implementation done by one layer can be modified by the another layer.

Elements of Layered Architecture

There are three different types of elements of a layered architecture. They are described below:

- **Service:** Service is defined as a set of functions and tasks being provided by a lower layer to a higher layer. Each layer performs a different type of task. Therefore, actions provided by each layer are different.
- **Protocol:** Protocol is defined as a set rules used by the layer for exchanging and transmission of data with its peer entities. This rules can consists details regarding a type of content and their order passed from one layer to another.
- **Interface:** Interface is defined as a channel that allows to transmit the messages from one layer to the another.

Need of Layered Architecture

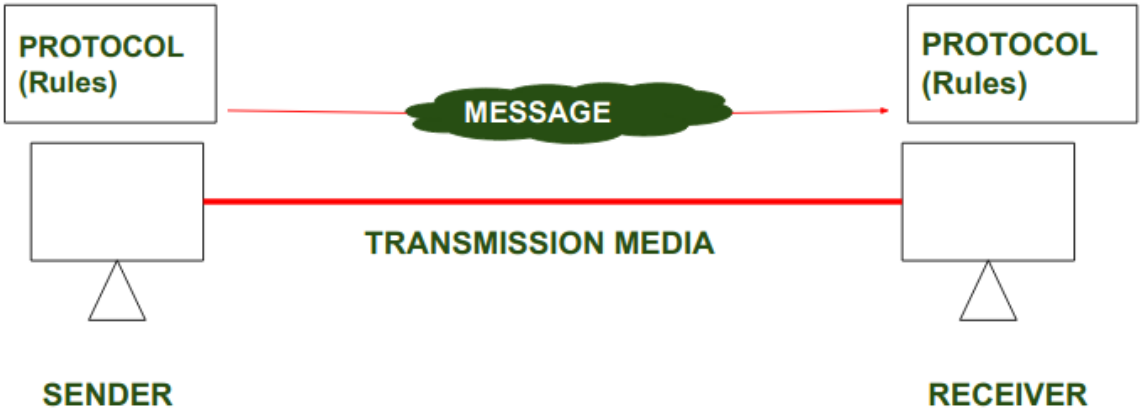
- **Divide and Conquer Approach:** Layered architecture supports divide and conquer approach. The unmanageable and complex task is further divided into smaller sub tasks. Each sub task is then carried out by the different layer. Therefore using this approach reduces the complexity of the problem or design process.
- **Easy to Modify:** The layers are independent of each other in layered architecture. If any sudden change occurs in the implementation of one layer, it can be changed. This change does not affect the working of other layers involved in the task. Therefore layered architectures are required to perform any sudden update or change.
- **Modularity:** Layered architecture is more modular as compared to other architecture models in computer network. Modularity provides with more independence between the layers and are more easier to understand.
- **Easy to Test:** Each layer in layered architecture performs a different and dedicated task. Therefore each layer can be analyzed and tested individually. It helps to analyze the problem and solve them more efficiently as compared to solving all the problems at a time.

Protocol and Standard in Computer Networks

Computer networks are dependent on protocols and standards which plays a vital role, which enables communication between different devices and systems with one another and share data seamlessly. Network protocol ensures that different technologies and components of the network are compatible with one another, reliable, and able to function together.

Overview of Protocol

In Order to make communication successful between devices, some rules and procedures should be agreed upon at the sending and receiving ends of the system. Such rules and procedures are called as Protocols. Different types of protocols are used for different types of communication.



In above diagrams Protocols are shown as set of rules. Such that Communication between Sender and Receiver is not possible without Protocol**.**
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Key Element of Protocol

- **Syntax:** syntax refers to the structure or the format of the data that gets exchanged between the devices. Syntax of message includes the type of data, composition of message and sequencing of message. The starting 8 bits of data is considered as the address of the sender. The next 8 bits is considered to be the address of the receiver. The remaining bits are considered as the message itself.
- **Semantics:** Semantics defines data transmitted between devices. It provides rules and norms for understanding message or data element values and actions.
- **Timing:** Timing refers to the synchronization and coordination between devices while transferring the data. Timing ensures at what time data should be sent and how fast data can be sent. For example, If a sender sends 100 Mbps but the receiver can only handle 1 Mbps, the receiver will overflow and lose data. Timing ensures preventing data loss, collisions and other timing related issues.
- **Sequence control:** Sequence control ensures the proper ordering of data packets. The main responsibility of sequence control is to acknowledge the data while it get received, and the retransmission of lost data. Through this mechanism the data is delivered in correct order.
- **Flow Control:** Flow control regulates device data delivery. It limits the sender's data or asks the receiver if it's ready for more. Flow control prevents data congestion and loss.
- **Error Control:** Error control mechanisms detect and fix data transmission faults. They include error detection codes, data resend, and error recovery. Error control detects and corrects noise, interference, and other problems to maintain data integrity.
- **Security:** Network security safeguards data confidentiality, integrity, and authenticity. which includes encryption, authentication, access control, and other security procedures. Network communication's privacy and trustworthiness are protected by security standards.

Standards

Standards are the set of rules for data communication that are needed for exchange of information among devices. It is important to follow Standards which are created by various Standard Organization like IEEE , ISO , ANSI etc.

Types of Standards

Standards are of two types :

- De Facto Standard.
- De Jure Standard.

De Facto Standard: The meaning of the work" *De Facto* " is " By Fact " or "By Convention".These are the standard s that have not been approved by any Organization , but have been adopted as Standards because of it's widespread use. Also, sometimes these standards are often established by Manufacturers.

For example: Apple and Google are two companies which established their own rules on their products which are different . Also they use some same standard rules for manufacturing for their products.

De Jure Standard: The meaning of the word *"De Jure"* is "By Law" or "By Regulations".Thus , these are the standards that have been approved by officially recognized body like ANSI , ISO , IEEE etc. These are the standard which are important to follow if it is required or needed.

For example : All the data communication standard protocols like SMTP, TCP , IP , UDP etc. are important to follow the same when we needed them.

Types of Protocol

- **Network Layer Protocols:** Network layer protocols operate in the network layer which is also known as the Layer 3 of the network architecture. Network layer protocols are responsible for packet routing, forwarding and addressing of data packets throughout the network. IP and ICMP are the network layer protocols.
- **Transport layer Protocols**: Transport layer protocols works in transport layer which provides end-to-end service ensuring data transfer across apps on different devices. TCP and UDP are the most popular transport layer protocols.
- **Application Layer Protocol :** Application layer protocol working in the application layer of the network architecture provides communication between applications running on different devices. The application layer protocols enable cross-device communication. They format, exchange, and interpret application data. HTTP, FTP, and SMTP are examples.
- **Wireless Protocols:** Wireless protocols basically used in wireless communication which enables data transfer through wireless networks. Bluetooth, Wi-Fi, and LTE protocols are examples.
- **Routing Protocols:** Routing protocol establishes the best/optimal network pathways throughout the network for fastest data transmission. Routers share information to develop and maintain routing tables. RIP, OSPF, and BGP are examples.
- **Security Protocols:** security protocol protects data confidentiality, integrity, and authenticity while transmission of data over the network. They include SSL and TLS, encryption methods, and authentication protocols for providing data security.
- **Internet Protocols:** IP identifies devices uniquely. Internet protocol provides data communication through routing and forwarding data packets from one device to another by unique addressing scheme.

Protocol and Standard Compliance in Network Security

Protocol and standard compliance protects data, resources, and networks. Protocol and standard compliance are crucial to network security for these reasons:

- **Interoperability:** Protocols and standards allow devices and systems to communicate. These protocols ensure network components can function together, avoiding risks and security gaps produced by incompatible or unsupported systems.
- **Security Baseline:** Protocols and standards contain security principles and best practices that help secure network infrastructure. These protocols allow organizations to protect sensitive data via encryption, authentication, and access controls.
- **Vulnerability Management:** Network security protocols and standards help organizations find and fix vulnerabilities. Many standards requires regular security assessments, vulnerability scanning, and penetration testing to discover network infrastructure flaws. Organizations can prevent cyberattacks and address vulnerabilities by following these compliance criteria.

Application of Layered Architecture in computer Networks

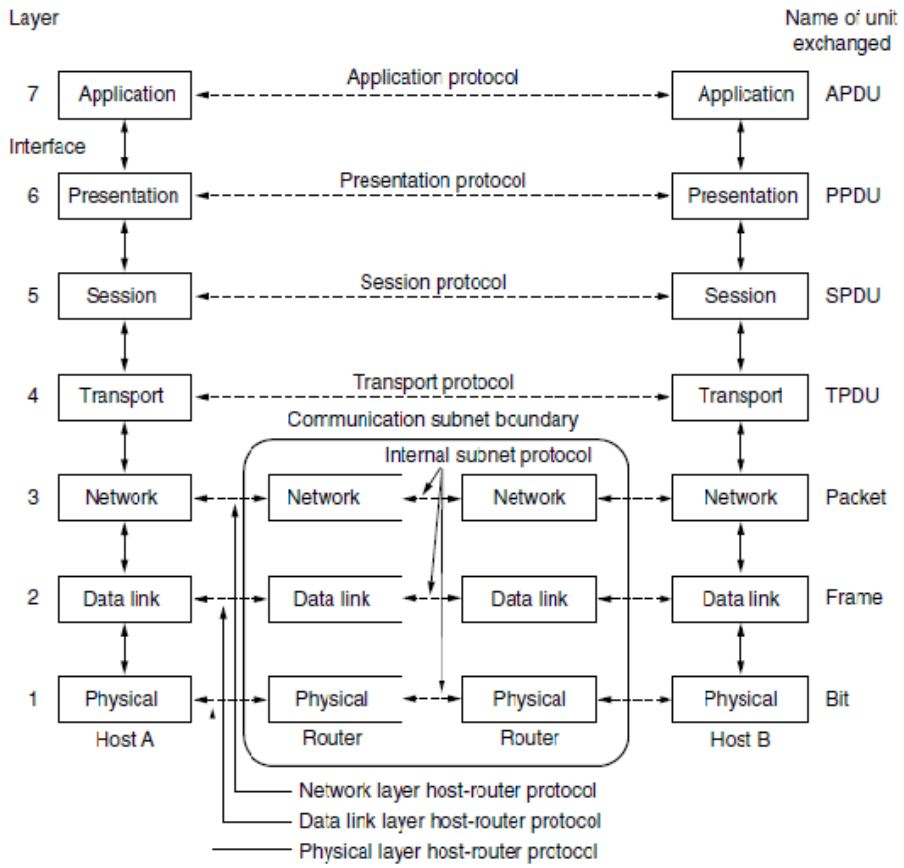
In computer networks, layered architecture is majorly used for communication. The two network models that makes use of layered architecture are:

- OSI Model
- TCP/IP Model

❖ OSI Reference Model

What is OSI Model?

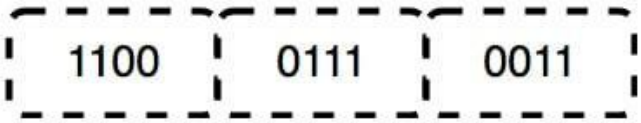
The OSI model, created in 1984 by ISO, is a reference framework that explains the process of transmitting data between computers. It is divided into seven layers that work together to carry out specialised network functions, allowing for a more systematic approach to networking.



What are the 7 layers of the OSI Model?
The OSI model consists of seven abstraction layers arranged in a top-down order:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Physical Layer – Layer 1
The lowest layer of the OSI reference model is the physical layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of **bits.** It is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.



Functions of the Physical Layer

- **Bit synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit rate control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus, star, or mesh topology.
- **Transmission mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full-duplex.

Note:
1. Hub, Repeater, Modem, and Cables are Physical Layer devices.
2. Network Layer, Data Link Layer, and Physical Layer are also known as **Lower Layers** or **Hardware Layers**.

Data Link Layer (DLL) – Layer 2
The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.
The Data Link Layer is divided into two sublayers:

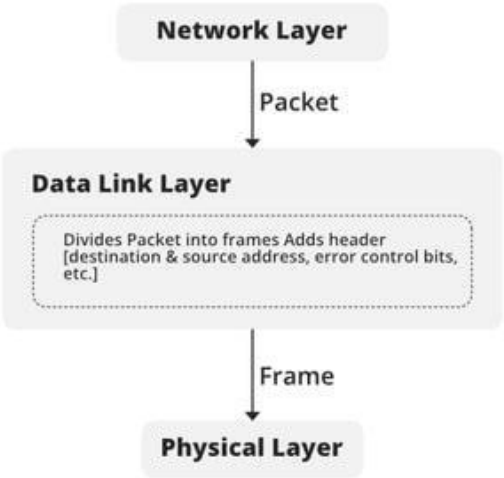
1. Logical Link Control (LLC)
2. Media Access Control (MAC)

The packet received from the Network layer is further divided into frames depending on the frame size of the NIC (Network Interface Card). DLL also encapsulates Sender and Receiver's MAC address in the header.

The Receiver's MAC address is obtained by placing an ARP (Address Resolution Protocol) request onto the wire asking "Who has that IP address?" and the destination host will reply with its MAC address.

Functions of the Data Link Layer

- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- **Physical addressing:** After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- **Error control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.



Note:

1. *Packet in the Data Link layer is referred to as **Frame.***
2. *Data Link layer is handled by the NIC (Network Interface Card) and device drivers of host machines.*
3. *Switch & Bridge are Data Link Layer devices.*

Network Layer – Layer 3

The network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender & receiver's IP addresses are placed in the header by the network layer.

Functions of the Network Layer

- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device on Internetwork uniquely, the network layer defines an addressing scheme. The sender & receiver's IP addresses are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.

Note:

1. *Segment in the Network layer is referred to as **Packet**.*
2. *Network layer is implemented by networking devices such as routers and switches.*

Transport Layer – Layer 4

The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as *Segments*. It is responsible for the End to End Delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.

At the sender's side: The transport layer receives the formatted data from the upper layers, performs **Segmentation**, and also implements **Flow & Error control** to ensure proper data transmission. It also adds Source and Destination port numbers in its header and forwards the segmented data to the Network Layer.

Note: The sender needs to know the port number associated with the receiver's application. Generally, this destination port number is configured, either by default or manually. For example, when a web application requests a web server, it typically uses port number 80, because this is the default port assigned to web applications. Many applications have default ports assigned.

At the receiver's side: Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. It also performs sequencing and reassembling of the segmented data.

Functions of the Transport Layer

- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer, and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Services Provided by Transport Layer

1. Connection-Oriented Service
2. Connectionless Service

1. Connection-Oriented Service: It is a three-phase process that includes

- Connection Establishment
- Data Transfer
- Termination/disconnection

In this type of transmission, the receiving device sends an acknowledgment, back to the source after a packet or group of packets is received. This type of transmission is reliable and secure.

2. Connectionless service: It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows for much faster communication between devices. Connection-oriented service is more reliable than connectionless Service.

Note:
1. Data in the Transport Layer is called **Segments**.
2. Transport layer is operated by the Operating System. It is a part of the OS and communicates with the Application Layer by making system calls.
3. The transport layer is called as **Heart of the OSI** model.
4. **Device or Protocol Use:** TCP, UDP NetBIOS, PPTP

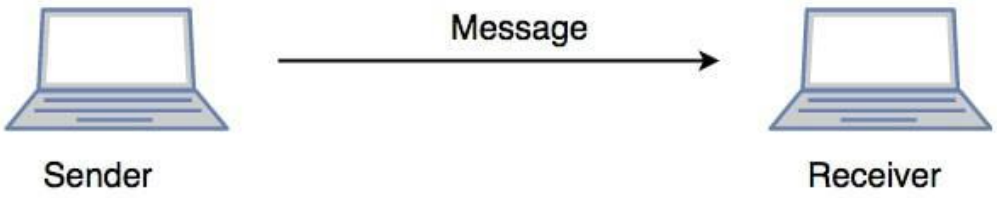
Session Layer – Layer 5
This layer is responsible for the establishment of connection, maintenance of sessions, and authentication, and also ensures security.

Functions of the Session Layer

- **Session establishment, maintenance, and termination:** The layer allows the two processes to establish, use and terminate a connection.
- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full-duplex.

Note:
1. *All the below 3 layers (including Session Layer) are integrated as a single layer in the TCP/IP model as the ???? pplication Layer".*
2. *Implementation of these 3 layers is done by the network application itself. These are also known as **Upper Layers or Software Layers.***
3. ***Device or Protocol Us :** NetBIOS, PPTP.*

for Example:
Let us consider a scenario where a user wants to send a message through some Messenger application running in his browser. The "Messenger" here acts as the application layer which provides the user with an interface to create the data. This message or so-called Data is compressed, encrypted (if any secure data), and converted into bits (0's and 1's) so that it can be transmitted.



Communication in Session Layer

Presentation Layer – Layer 6
The presentation layer is also called the **Translation layer**. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.

Functions of the Presentation Layer

- **Translation:** For example, ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the cipher text and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

Note: **Device or Protocol Use:** JPEG, MPEG, GIF

Application Layer – Layer 7

At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data, which has to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user.

Example: Application – Browsers, Skype Messenger, etc.

Note: 1. The application Layer is also called Desktop Layer.

2. **Device or Protocol Use:** SMTP

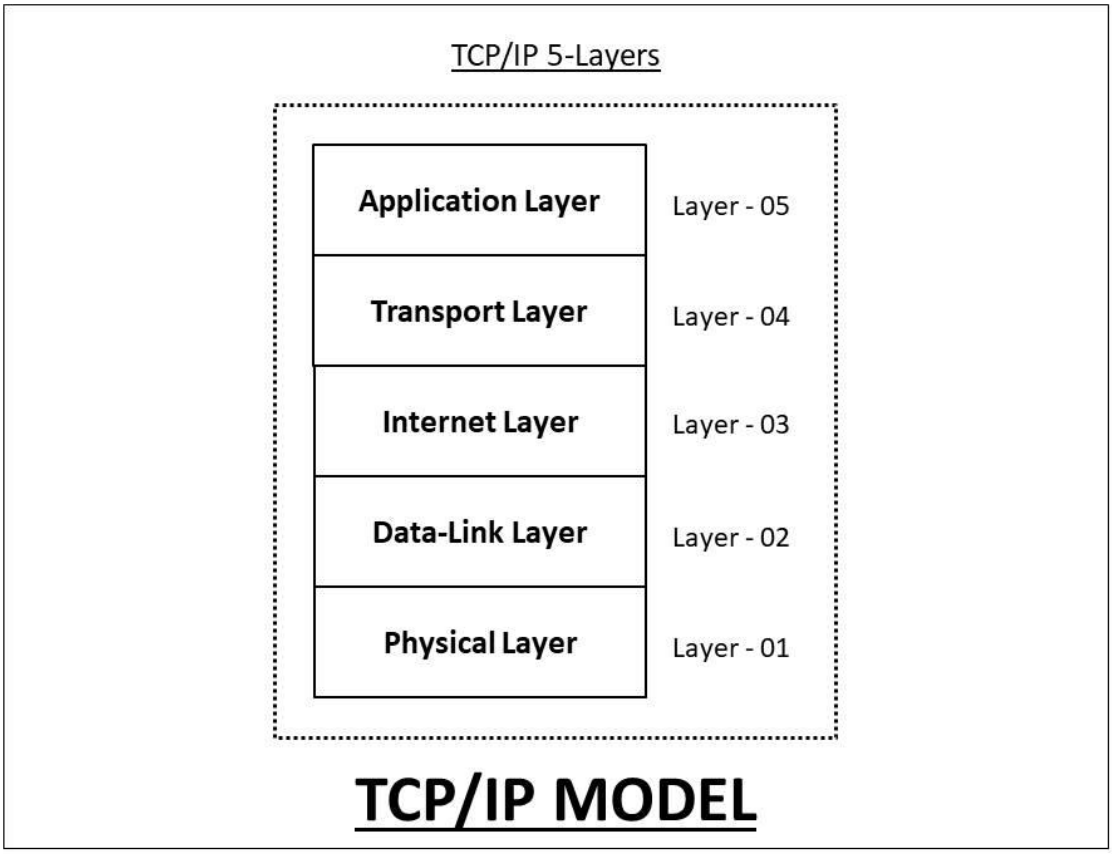
Functions of the Application Layer

The main functions of application layer are given below.

- Network Virtual Terminal: It allows a user to log on to a remote host.
- FTAM- File transfer access and management : This application allows a user to access file in a remote host, retrieve files in remote host and manage or control files from a remote computer.
- Mail Services: Provide email service.
- Directory Services : This application provides distributed database sources and access for global information about various objects and services.

❖ TCP/IP Protocol Suite

TCP and IP are different protocols of Computer Networks. The basic difference between TCP (Transmission Control Protocol) and IP (Internet Protocol) is in the transmission of data. In simple words, IP finds the destination of the mail and TCP has the work to send and receive the mail. UDP is another protocol, which does not require IP to communicate with another computer. IP is required by only TCP. This is the basic difference between TCP and IP.



1. Physical Layer

It is a group of applications requiring network communications. This layer is responsible for generating the data and requesting connections. It acts on behalf of the sender and the Network Access layer on the behalf of the receiver. During this article, we will be talking on the behalf of the receiver.

2. Data Link Layer

The packet's network protocol type, in this case, TCP/IP, is identified by the data-link layer. Error prevention and "framing" are also provided by the data-link layer. Point-to-Point Protocol (PPP) framing and Ethernet IEEE 802.2 framing are two examples of data-link layer protocols.

3. Internet Layer

This layer parallels the functions of OSI's Network layer. It defines the protocols which are responsible for the logical transmission of data over the entire network. The main protocols residing at this layer are as follows:

- **IP:** IP stands for Internet Protocol and it is responsible for delivering packets from the source host to the destination host by looking at the IP addresses in the packet headers. IP has 2 versions: IPv4 and IPv6. IPv4 is the one that most websites are using currently. But IPv6 is growing as the number of IPv4 addresses is limited in number when compared to the number of users.

- **ICMP:** ICMP stands for Internet Control Message Protocol. It is encapsulated within IP datagrams and is responsible for providing hosts with information about network problems.
- **ARP:** ARP stands for Address Resolution Protocol. Its job is to find the hardware address of a host from a known IP address. ARP has several types: Reverse ARP, Proxy ARP, Gratuitous ARP, and Inverse ARP.

The Internet Layer is a layer in the Internet Protocol (IP) suite, which is the set of protocols that define the Internet. The Internet Layer is responsible for routing packets of data from one device to another across a network. It does this by assigning each device a unique IP address, which is used to identify the device and determine the route that packets should take to reach it.

Example: Imagine that you are using a computer to send an email to a friend. When you click "send," the email is broken down into smaller packets of data, which are then sent to the Internet Layer for routing. The Internet Layer assigns an IP address to each packet and uses routing tables to determine the best route for the packet to take to reach its destination. The packet is then forwarded to the next hop on its route until it reaches its destination. When all of the packets have been delivered, your friend's computer can reassemble them into the original email message.

In this example, the Internet Layer plays a crucial role in delivering the email from your computer to your friend's computer. It uses IP addresses and routing tables to determine the best route for the packets to take, and it ensures that the packets are delivered to the correct destination. Without the Internet Layer, it would not be possible to send data across the Internet.

4. Transport Layer

The TCP/IP transport layer protocols exchange data receipt acknowledgments and retransmit missing packets to ensure that packets arrive in order and without error. End-to-end communication is referred to as such. Transmission Control Protocol (TCP) and User Datagram Protocol are transport layer protocols at this level (UDP).

- **TCP:** Applications can interact with one another using TCP as though they were physically connected by a circuit. TCP transmits data in a way that resembles character-by-character transmission rather than separate packets. A starting point that establishes the connection, the whole transmission in byte order, and an ending point that closes the connection make up this transmission.
- **UDP:** The datagram delivery service is provided by UDP, the other transport layer protocol. Connections between receiving and sending hosts are not verified by UDP. Applications that transport little amounts of data use UDP rather than TCP because it eliminates the processes of establishing and validating connections.

5. Application Layer

This layer is analogous to the transport layer of the OSI model. It is responsible for end-to-end communication and error-free delivery of data. It shields the upper-layer applications from the complexities of data. The three main protocols present in this layer are:

- **HTTP and HTTPS:** HTTP stands for Hypertext transfer protocol. It is used by the World Wide Web to manage communications between web browsers and servers. HTTPS stands for HTTP-Secure. It is a combination of HTTP with SSL (Secure Socket Layer). It is efficient in cases where the browser needs to fill out forms, sign in, authenticate, and carry out bank transactions.
- **SSH:** SSH stands for Secure Shell. It is a terminal emulations software similar to Telnet. The reason SSH is preferred is because of its ability to maintain the encrypted connection. It sets up a secure session over a TCP/IP connection.
- **NTP:** NTP stands for Network Time Protocol. It is used to synchronize the clocks on our computer to one standard time source. It is very useful in situations like bank transactions. Assume the following situation without the presence of NTP. Suppose you carry out a transaction, where your computer reads the time at 2:30 PM while the server records it at 2:28 PM. The server can crash very badly if it's out of sync.

The host-to-host layer is a layer in the OSI (Open Systems Interconnection) model that is responsible for providing communication between hosts (computers or other devices) on a network. It is also known as the transport layer.
Some common use cases for the host-to-host layer include:

1. **Reliable Data Transfer:** The host-to-host layer ensures that data is transferred reliably between hosts by using techniques like error correction and flow control. For example, if a packet of data is lost during transmission, the host-to-host layer can request that the packet be retransmitted to ensure that all data is received correctly.
2. **Segmentation and Reassembly:** The host-to-host layer is responsible for breaking up large blocks of data into smaller segments that can be transmitted over the network, and then reassembling the data at the destination. This allows data to be transmitted more efficiently and helps to avoid overloading the network.
3. **Multiplexing and Demultiplexing:** The host-to-host layer is responsible for multiplexing data from multiple sources onto a single network connection, and then demultiplexing the data at the destination. This allows multiple devices to share the same network connection and helps to improve the utilization of the network.
4. **End-to-End Communication:** The host-to-host layer provides a connection-oriented service that allows hosts to communicate with each other end-to-end, without the need for intermediate devices to be involved in the communication.

- **Network Devices**

Network devices, also known as networking hardware, are physical devices that allow hardware on a computer network to communicate and interact with one another. For example Repeater, Hub, Bridge, Switch, Routers, Gateway, Brouter, and NIC, etc.

1. Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they not only amplify the signal but also regenerate it. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength. It is a 2-portdevice.

2. Hub – A hub is a basically multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

Types of Hub

- **Active Hub:-** These are the hubs that have their power supply and can clean, boost, and relay the signal along with the network. It serves both as a repeater as well as a wiring center. These are used to extend the maximum distance between nodes.
- **Passive Hub:-** These are the hubs that collect wiring from nodes and power supply from the active hub. These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend the distance between nodes.
- **Intelligent Hub:-** It works like an active hub and includes remote management capabilities. They also provide flexible data rates to network devices. It also enables an administrator to monitor the traffic passing through the hub and to configure each port in the hub.

3. Bridge – A bridge operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

Types of Bridges

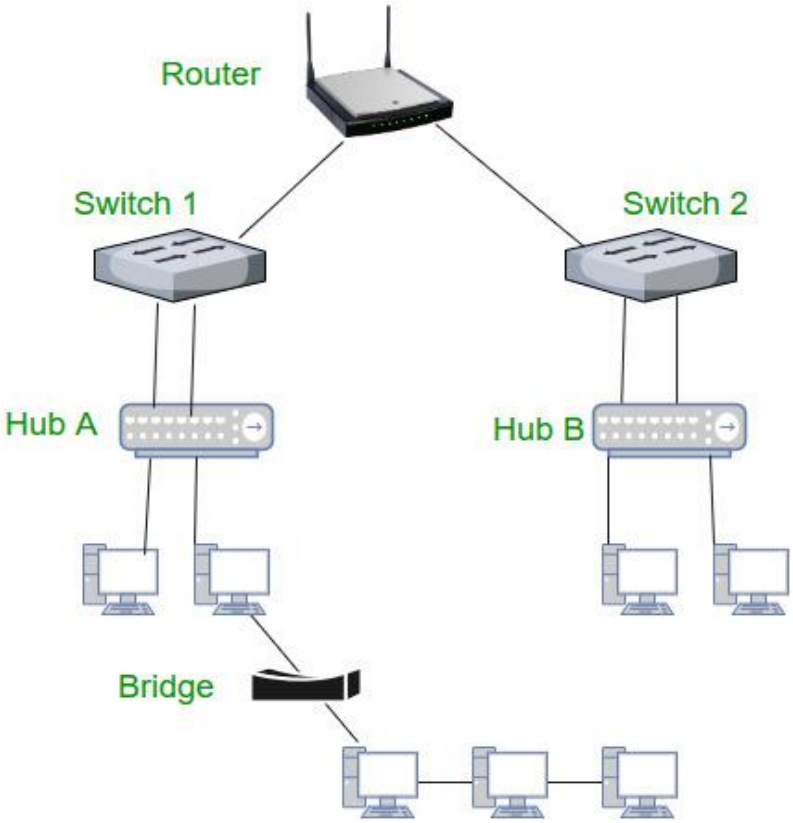
- **Transparent Bridges:-** These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.
- **Source Routing Bridges:-** In these bridges, routing operation is performed by the source station and the frame specifies which route to follow. The host can discover the frame by sending a special frame called the discovery frame, which spreads through the entire network using all possible paths to the destination.

4. Switch – A switch is a multiport bridge with a buffer and a design that can boost its efficiency (a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but the broadcast domain remains the same.

Types of Switch

1. Unmanaged switches: These switches have a simple plug-and-play design and do not offer advanced configuration options. They are suitable for small networks or for use as an expansion to a larger network.
2. Managed switches: These switches offer advanced configuration options such as VLANs, QoS, and link aggregation. They are suitable for larger, more complex networks and allow for centralized management.
3. Smart switches: These switches have features similar to managed switches but are typically easier to set up and manage. They are suitable for small- to medium-sized networks.
4. Layer 2 switches: These switches operate at the Data Link layer of the OSI model and are responsible for forwarding data between devices on the same network segment.
5. Layer 3 switches: These switches operate at the Network layer of the OSI model and can route data between different network segments. They are more advanced than Layer 2 switches and are often used in larger, more complex networks.
6. PoE switches: These switches have Power over Ethernet capabilities, which allows them to supply power to network devices over the same cable that carries data.
7. Gigabit switches: These switches support Gigabit Ethernet speeds, which are faster than traditional Ethernet speeds.
8. Rack-mounted switches: These switches are designed to be mounted in a server rack and are suitable for use in data centers or other large networks.
9. Desktop switches: These switches are designed for use on a desktop or in a small office environment and are typically smaller in size than rack-mounted switches.
10. Modular switches: These switches have modular design, which allows for easy expansion or customization. They are suitable for large networks and data centers.

5. Routers – A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.



6. Gateway – A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

7. Brouter – It is also known as the bridging router is a device that combines features of both bridge and router. It can work either at the data link layer or a network layer. Working as a router, it is capable of routing packets across networks and working as the bridge, it is capable of filtering local area network traffic.

8. NIC – NIC or network interface card is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique id that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and the router or modem. NIC card is a layer 2 device which means that it works on both the physical and data link layers of the network model.

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